

- 2 The veins of a leaf contain transport tissues. Fig. 2.1 is a drawing made from an electron micrograph showing a cross-section of the transport tissue in a leaf vein.

The cells labelled **A** are modified companion cells, known as phloem transfer cells. Transfer cells move sucrose and other assimilates from mesophyll cells, **B**, into the phloem sieve tube element, **C**.

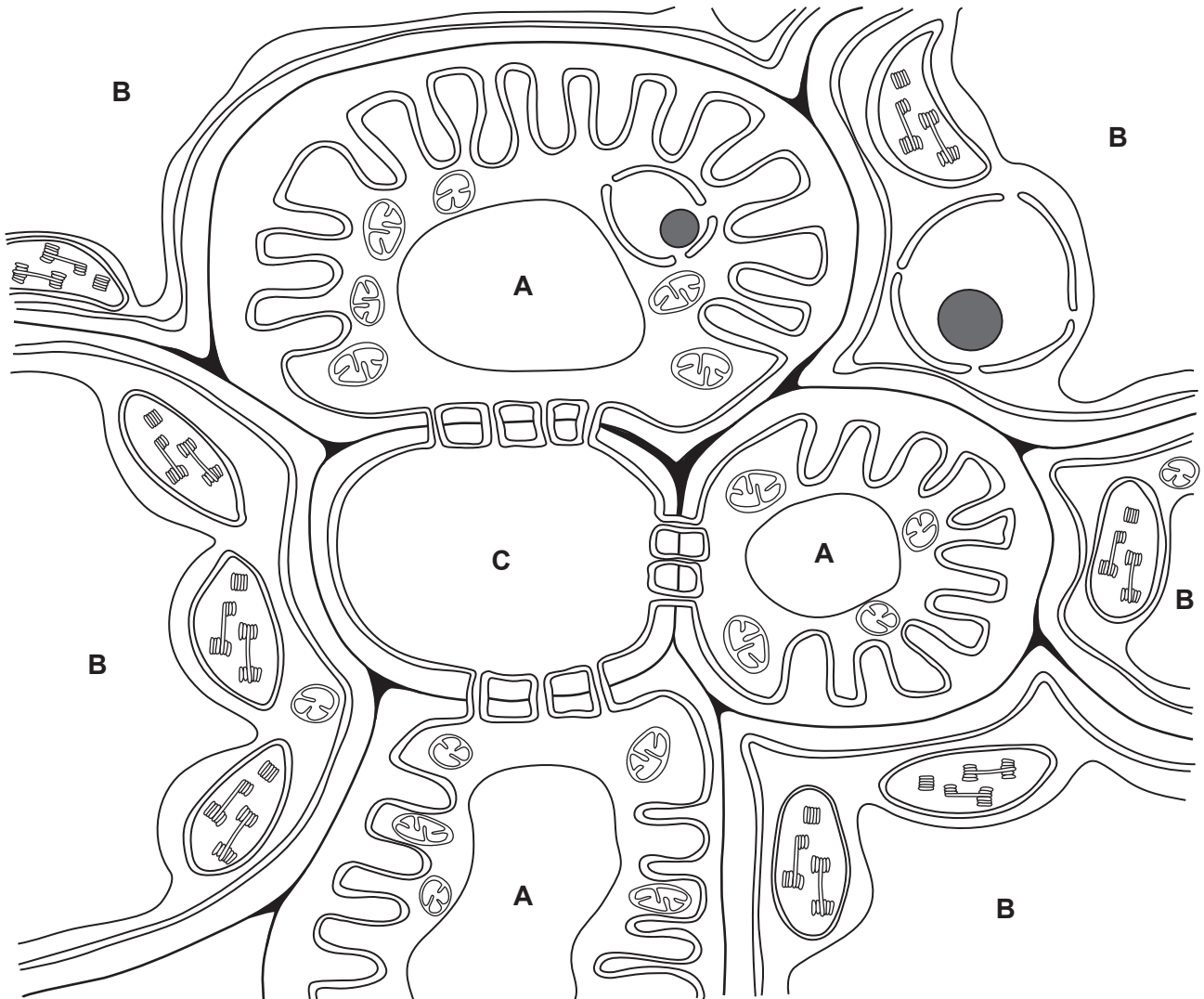


Fig. 2.1

- (a) State the functions of the mesophyll cells, **B**, and sieve tube element, **C**, and explain how their structure is adapted for their function.

cell **B** – function

adaptation

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cell **C** – function

adaptation

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- (b) The cell walls of the transfer cells, **A**, shown in Fig. 2.1, have infoldings.

Explain the advantages of these cell wall infoldings for the movement of sucrose from mesophyll cells to phloem sieve tubes.

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